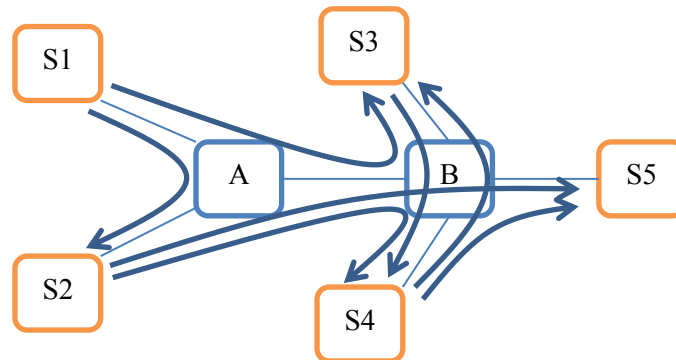


Application of the performance analysis methodology of spacewire



Let's consider the illustrated topology of spacewire network with the following flows :

f1 : S1 => A => B => S3 , L(f1) = 200bytes, P = 20ms

f2 : S1 => A => S2, L(f2) = 200bytes, P = 20ms

f3 : S2 => A => B => S5, L(f3) = 200bytes, P = 20ms

f4 : S2 => A => B => S4 , L(f4) = 200bytes, P = 20ms

f5 : S3 => B => S4, L(f5) = 600bytes, P = 20ms

f6 : S4 => B => S5, L(f6) = 200bytes, P = 20ms

f7 : S4 => B => S3, L(f7) = 400bytes, P = 20ms

The transmission capacity of the spacewire is 10Mbps.

1. What is the worst-case end-to-end delay of flow f1 ?
2. What is the worst-case end-to-end delay of flow f4 ?
3. What is the worst-case end-to-end delay of flow f7 ?